Functional Software Documentation (FSD)

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Program	SkillWallet, ServiceNow Global Certification
Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 6: Project Documentation - Functional Software Documentation (FSD)

1. Introduction

Project Title: Car Rental Request Automation in ServiceNow
Primary Developer & System Administrator: Nandini Bade
Program: SkillWallet, ServiceNow Global Certification

2. Project Overview

Purpose: Automate corporate vehicle rental requests via ServiceNow Service Catalog, Flow Designer, and Business Rules, replacing legacy paper forms and unmonitored email communications.
Key Features: Self-service catalog request form, dynamic vehicle choice selection, automated approval workflows, Transport Team catalog task assignment, automated email dispatch, and exception incident generation for unavailable vehicles.

3. System Architecture

Frontend: Service Portal Service Catalog (sc_cat_item).
Process & Flow Engine: Workflow Studio / Flow Designer (sys_hub_flow).
Backend Database: ServiceNow MariaDB relational tables (sc_request, sc_req_item, sc_task, incident, sys_user).
Server-Side Logic: Business Rules using GlideRecord API.

4. Setup & Instance Configuration

Prerequisites: ServiceNow Developer Instance (Vancouver/Washington DC release) with admin rights.
Configuration Order:
1. Create Update Set
2. Maintain Item 'Car Rental Request'
3. Add 7 Variables & Choice list (Mini, SUV, Sedan)
4. Build & Activate Flow 'Car Rental Req'

5. Configure Business Rule 'Create Incident If Car Unavailable'
6. Attach Flow to Catalog Item Process Engine

5. Data Dictionary & Table Structure

Table 1: sc_cat_item (Catalog Item Record)
Table 2: sc_req_item (Requested Item Data Store)
Table 3: sc_task (Fulfillment Catalog Task assigned to Transport Team)
Table 4: incident (Exception Incident Ticket created on State=3)
Table 5: sys_email (Email Log Store)

6. Execution Workflow Specifications

Trigger: Service Catalog item submission.
Step 1: Ask For Approval (Record: sc_req_item, Rule: Approve when Anyone approves).
Step 2: If Approval State = Approved.
Step 3: Create Catalog Task (Table: sc_task, Group: Transport Team, Description: 'Verify This Item And Assign To A Person Who Manages', Wait: true).
Step 4: Get Catalog Variables.
Step 5: Send Email to Requestor.

7. API & GlideRecord Script Specifications

Business Rule Script (Table: sc_req_item, When: After Insert/Update, Order: 100):

```
(function executeRule(current, previous) {
    if (current.state == 3) {
        var inc = new GlideRecord('incident');
        inc.initialize();
        inc.short_description = 'Car unavailable for request: ' + current.number;
        inc.description = 'Car not available for trip on ' + current.variables.requested_date;
        inc.insert();
    }
})(current, previous);
```

8. Security & Access Control (ACLs)

Role-based Access Control (RBAC):
• Requestor: Submit catalog requests, view personal RITM status.
• Approver: Review & decide on approval records.
• Transport Team: View & close assigned catalog tasks (sc_task).
• Admin (Nandini Bade): Full configuration access.

9. User Interface & Form Layout

Item Name: Car Rental Request
Category: Transportation Services
Fields: Car Type (Select Box), Requested Date (Date), Pickup Location (Text), Duration Hours (Numeric Scale), Requestor (Reference), Drop Location (Text), Reason (Multi-line Text).

10. Testing Strategy & Execution

Test Suite covering 11 Test Cases (T1-T9 Functional, T10-T11 Negative/Exception). 100% pass rate achieved on developer instance.

11. Verification & Output Audit

Confirmed REQ, RITM, sc_task, sys_email log, and incident records created cleanly with complete approval audit history.

12. Known Issues & Workarounds

Current static approver rule requires manual adjustment for multi-level manager hierarchy. Workaround: Use dynamic manager script in future release.

13. Future Enhancements

- Dynamic Manager Approval Routing.
- Automated SLA Definitions on sc_task.
- Real-time Car Inventory Integration table.